

Sara Ghasvarianjahromi, PhD

ML Research Engineer | LLMs, Information Retrieval & Robust Machine Learning

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Summary

Ph.D. in Electrical Engineering specializing in large language models, semantic retrieval, and robust decentralized machine learning. Experience building retrieval systems, adversarially robust learning algorithms, and scalable machine learning systems, with additional applied work in wireless network optimization. Proficient in PyTorch, optimization, and scalable machine learning, with multiple peer-reviewed IEEE publications.

Skills

Machine Learning: Large Language Models (LLMs), Semantic Retrieval, Retrieval-Augmented Generation (RAG), Decentralized Learning, Adversarial Robustness, Deep Learning, AI Agents

Programming: Python, MATLAB

ML Frameworks & Libraries: PyTorch, Hugging Face Transformers, pandas, NumPy, scikit-learn

Optimization & Modeling: Gurobi, Convex Optimization, Statistical Modeling, Probabilistic Modeling

Tools: Git, Linux, VS Code, LaTeX

Experience

New Jersey Institute of Technology, NJ – Research Assistant

Jan 2022 – May 2026

- Built a context-aware semantic retrieval system using pretrained embeddings to improve retrieval accuracy under noisy and lossy communication conditions.
- Developed and validated decentralized ML architectures with adversarial defenses in PyTorch, achieving 30–50% higher accuracy than state-of-the-art baselines and up to 25% improvement under adversarial noise on MNIST and CIFAR-10.
- Implemented a fully decentralized sparse matrix multiplication framework with formal robustness guarantees, enabling scalable coordination-free distributed computation.
- Derived probabilistic performance bounds to characterize system behavior under uncertainty.

Futurewei Technologies, NJ – ML Research Intern

May 2025 – Aug 2025

- Built an end-to-end AI/ML pipeline for QoS optimization using 3GPP network analytics, reducing energy consumption by 18–25% over fixed-latency baselines and 40–50% over non-energy-aware systems while reducing latency by 30–40%.
- Trained neural network models in PyTorch to infer adaptive, per-user QoS recommendations from network telemetry, replacing rule-based decision logic with data-driven inference.
- Designed a Gurobi-based optimization pipeline for energy-aware path selection and transmit-power control, jointly minimizing latency and energy under strict QoS constraints.

Ozyegin University, Istanbul, Turkey – Research Assistant

Jan 2018 – Jul 2021

- Developed convex optimization algorithms for energy-efficient wireless systems, extending device battery lifetime by up to 110%.

Projects

AI Job Scout Agent

- Built a multi-agent AI job search assistant integrating LLM reasoning, semantic ranking, SQLite persistence, Git-based development, and automated cover letter/LinkedIn message generation.
- Developed in Python using the Gemini API, modular agent architecture, and multiple job data sources.

Parameter-Efficient LLM Fine-Tuning

- Fine-tuned DistilBERT using LoRA (PEFT) on SST-2 sentiment classification with Hugging Face Transformers and PyTorch.

Education

PhD in Electrical Engineering, New Jersey Institute of Technology

Newark, NJ

Jan 2022 – May 2026

Dissertation: Reliable Machine Learning and Information Retrieval under Uncertainty

Certifications & Professional Development

Google AI Agents Workshop – Built and deployed LLM-based agentic AI applications using Google’s Agent Development Kit (ADK) and Vertex AI during an intensive hands-on workshop. (Google, Jun 2026)

Google Machine Learning Crash Course ([badges](#)) – LLMs, embeddings, and production ML systems modules (Google, 2025–2026)

Selected Publications

Full list on [Google Scholar](#).

1. **S. Ghasvarianjahromi**, J. Barr, Y. Yakimenka, J. Kliewer, “Context-Aware Search and Retrieval Under Token Erasure,” Submitted, 2026.
2. **S. Ghasvarianjahromi**, A. Kiani, A. Xiang, J. Kaippallimalil, T. Saboorian, N. Ansari, “AI/ML-based QoS Recommendations for Energy Optimization in 6G Networks,” *IEEE Netw. Lett.*, 2025.
3. M. Bakshi, **S. Ghasvarianjahromi**, Y. Yakimenka, A. Beemer, O. Kosut, J. Kliewer, “VALID: A Validated Algorithm for Learning in Decentralized Networks with Possible Adversarial Presence,” *IEEE Intern. Symp. Info. Theo. (ISIT)*, 2024.

Professional Service

IEEE Reviewer – Wireless Communications Letters, Communications Letters, Transactions on Communications, Transactions on Vehicular Technology, Transactions on Wireless Communications, Elsevier Physical Communications